

Уравнения Гамильтона - Якоби

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases}$$

$i = \overline{1, n}$

в классе свободных унитарн. ($c=1$) преобр.

$$\tilde{q} = \tilde{q}(q, p, t)$$

$$\tilde{p} = \tilde{p}(q, p, t)$$

$$\Rightarrow \begin{cases} \dot{\tilde{q}}_i = \frac{\partial \tilde{H}}{\partial \tilde{p}_i} \\ \dot{\tilde{p}}_i = -\frac{\partial \tilde{H}}{\partial \tilde{q}_i} \end{cases}$$

$$\text{Если } \tilde{q}, \tilde{p} : \tilde{H} = 0$$

$$\begin{cases} \dot{\tilde{q}}_i = 0 \\ \dot{\tilde{p}}_i = 0 \end{cases} \Rightarrow \begin{cases} \tilde{q}_i = \alpha_i \\ \tilde{p}_i = \beta_i \end{cases} \Rightarrow \begin{cases} q_i = q_i(\bar{\alpha}, \bar{\beta}, t) \\ p_i = p_i(\bar{\alpha}, \bar{\beta}, t) \end{cases}$$

$$\begin{cases} \frac{\partial S}{\partial q_i} = p_i \\ \frac{\partial S}{\partial \tilde{q}_i} = -\tilde{p}_i \end{cases}$$

$$\tilde{H} = cH + \frac{\partial S}{\partial t}$$

Как найти преобразование?

$$\tilde{H} = 0$$

$$S = S(q, \tilde{q}, t)$$

$$0 = H + \frac{\partial S}{\partial t}$$

Уравнение Гамильтона - Якоби:

$$H(q, \frac{\partial S}{\partial q}, t) + \frac{\partial S}{\partial t} = 0 \quad (*)$$

Опр.

$S(q, \alpha, t)$ - полный интеграл, если:

- S -реш. УГЯ (*)

$$- \det \left| \frac{\partial^2 S}{\partial q_i \partial \alpha_j} \right| \neq 0$$

th Якоби

Если дана S -полный интеграл, то конечные ур-я движения c -мн можно получить из соотнош.:

$$\frac{\partial S}{\partial q_i} = p_i$$

$$\frac{\partial S}{\partial \alpha_i} = -\beta_i$$

$i = \overline{1, n}$

$$\Delta. S = S(\bar{q}, \bar{\alpha}, t)$$

$$\frac{\partial S}{\partial q_i} = f_i(\bar{q}, \bar{\alpha}, t) = p_i \quad i = \overline{1, n}$$

$$\frac{\partial S}{\partial \alpha_i} = g_i(\bar{q}, \bar{\alpha}, t) = -\beta_i \quad i = \overline{1, n}$$

$$\Downarrow \\ q_i = q_i(\bar{\alpha}, \bar{\beta}, t)$$

$$\left| \frac{\partial g_i}{\partial q_j} \right|_{i,j} \neq 0 \rightarrow \left| \frac{\partial^2 S}{\partial q_j \partial \alpha_i} \right| \neq 0$$



Частные случаи

1. Циклич. коорд.

$$\frac{\partial H}{\partial q_l} = 0, \quad l = \overline{k+1, n}$$

$$H(q_1, \dots, q_k, p_1, \dots, p_n, t)$$

$$\frac{\partial S}{\partial t} + H(q_1, \dots, q_k, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}, t) = 0$$

$$\frac{\partial H}{\partial q_l} = 0 \Rightarrow p_l = \alpha_l, \quad l = \overline{k+1, n}$$

$$p_l = \frac{\partial S}{\partial q_l} = \alpha_l$$

$$S = \sum_{l=k+1}^n \alpha_l q_l + S^*(q_1, \dots, q_k, \alpha_1, \dots, \alpha_n, t)$$

$$\frac{\partial S^*}{\partial t} + H(q_1, \dots, q_k, \frac{\partial S^*}{\partial q_1}, \dots, \frac{\partial S^*}{\partial q_k}, \alpha_{k+1}, \dots, \alpha_n, t) = 0$$

2. Обобщ. конс. сист

$$\frac{\partial H}{\partial t} = 0 \Rightarrow \frac{dH}{dt} = 0 \quad H - \text{ПУ}$$

$$H = h \quad S = S(q_1, \dots, q_n, \alpha_1, \dots, \alpha_{n-1}, \overset{\alpha_n}{h}, t)$$

$$\frac{\partial S}{\partial t} + H = 0 \Rightarrow \frac{\partial S}{\partial t} = -h$$

$$\Rightarrow S = -ht + V(\bar{q}, \bar{\alpha}) \quad \bar{\alpha} = (\alpha_1, \dots, \alpha_{n-1}, h)^T$$

$$H(\bar{q}, \frac{\partial V}{\partial \bar{q}}) = h \quad - \text{УГД} \quad \text{для } V \Rightarrow V(\bar{q}, \bar{\alpha}) \Rightarrow S$$

$$p_i = \frac{\partial S}{\partial q_i} = \frac{\partial V}{\partial q_i} \quad i = \overline{1, n}$$

$$-p_i = \frac{\partial S}{\partial \alpha_i} = \frac{\partial V}{\partial \alpha_i} \quad i = \overline{1, n-1}$$

$$-p_n = \frac{\partial S}{\partial \alpha_n} = -t + \frac{\partial V}{\partial \alpha_n} \quad i = n \quad (\alpha_n = h)$$

3. Метод разделения переменных

$$S(\bar{q}, \bar{\alpha}, t) = S_0(\bar{\alpha}, t) + \sum_{i=1}^n S_i(q_i, \bar{\alpha})$$

Част. случ.

$$a) H = H(f_1(q_1, p_1), f_2(q_2, p_2), \dots, t)$$

$$f_i = f_i(q_i, p_i) = \alpha_i \quad - \text{ПУ}$$

$$S = S_0(\bar{\alpha}, t) + \sum_{i=1}^n S_i(q_i, \bar{\alpha})$$

$$\frac{\partial S_0}{\partial t} + H(q_1, \dots, q_n, \frac{\partial S_1}{\partial q_1}, \dots, \frac{\partial S_n}{\partial q_n}, t) = 0$$

$$\frac{\partial S_0}{\partial t} + H(\alpha_1, \dots, \alpha_n, t) = 0$$

$$S_0 = - \int H(\alpha_1, \dots, \alpha_n, t) dt$$

$$S = - \int H(\alpha_1, \dots, \alpha_n, t) dt + \sum_{i=1}^n S_i(q_i, \bar{\alpha})$$

$$\int \frac{\partial f_i}{\partial p_i} \neq 0 \quad i = \overline{1, n}$$

$$\Leftrightarrow p_i = g_i(q_i, \alpha_i) = \frac{\partial S}{\partial q_i} = \frac{\partial S_i}{\partial q_i}$$

$$\Rightarrow S_i = \int g_i(q_i, \alpha_i) dq_i$$

$$S = - \int H(\alpha_1, \dots, \alpha_n, t) dt + \sum_{i=1}^n \int g_i(q_i, \alpha_i) dq_i$$

! полный интеграл !

$$\left| \frac{\partial^2 S}{\partial q_i \partial \alpha_j} \right| = \left| \frac{\partial g_i(q_i, \alpha_i)}{\partial \alpha_j} \right|_{i,j} = \det \begin{pmatrix} \frac{\partial g_1}{\partial \alpha_1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \frac{\partial g_n}{\partial \alpha_n} \end{pmatrix} = \prod_{i=1}^n \frac{\partial g_i}{\partial \alpha_i} = \prod_{i=1}^n \frac{1}{\frac{\partial f_i}{\partial p_i}}$$

$$f_i(q_i, p_i) = \alpha_i \quad \frac{\partial f_i}{\partial \alpha_i} = 1 = \frac{\partial f_i}{\partial p_i} \frac{\partial g_i}{\partial \alpha_i} \quad \frac{\partial g_i}{\partial \alpha_i} = 1 / \frac{\partial f_i}{\partial p_i}$$

$\uparrow$
 $g_i(q_i, \alpha_i)$

$$\delta) H = H(f_n(q_n, p_n, f_{n-1}(\dots)), t)$$

$$f_k(q_k, p_k, f_{k-1}) = f_k$$

$$f_1(q_1, p_1) = f_1$$

$$f_k(q_k, p_k, f_{k-1}) = \alpha_k$$

$$f_k(q_k, p_k, \alpha_{k-1}) = \alpha_k$$

$$S = S_0(\bar{\alpha}, t) + \sum_{i=1}^n S_i(q_i, \bar{\alpha})$$

$$\frac{\partial S_0}{\partial t} + H(\alpha_n, t) = 0$$

$$S_0 = - \int H(\alpha_n, t) dt$$

$$\int \frac{\partial f_k}{\partial p_k} \neq 0 \Rightarrow p_k = g_k(\alpha_k, \alpha_{k-1}, q_k)$$

$$p_k = \frac{\partial S}{\partial q_k} = \frac{\partial S_k}{\partial q_k} \Rightarrow S_k = \int g_k(\alpha_k, \alpha_{k-1}, q_k) dq_k$$

! что полн. инт.

$$\left| \frac{\partial^2 S}{\partial q_k \partial \alpha_j} \right|_{k,j} = \left| \frac{\partial g_k}{\partial \alpha_j} \right|_{k,j} = \det \begin{pmatrix} \frac{\partial g_1}{\partial \alpha_1} & \dots & 0 \\ \frac{\partial g_2}{\partial \alpha_1} & \frac{\partial g_2}{\partial \alpha_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} = \prod_{k=1}^n \frac{\partial g_k}{\partial \alpha_k} = \prod_{k=1}^n \frac{1}{\frac{f_k}{p_k}} \neq 0 !$$

$$S = S_0(\bar{\alpha}, t) + \sum_{i=1}^n S_i(q_i, \bar{\alpha})$$

$$\frac{\partial S_i}{\partial q_i} = p_i = g_i(\bar{q}, \bar{\alpha})$$

$$\Downarrow$$

откуда-то

прим.

$$\text{движение точки по прямой}$$

$$T = \frac{m \dot{x}^2}{2}, \quad \Pi = 0, \quad L = T - \Pi$$

$$x(0) = x_0, \quad \dot{x}(0) = \dot{x}_0 > 0$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m \dot{x} \Rightarrow \dot{x} = \frac{p_x}{m}$$

$$H = \frac{p_x^2}{2m} = h \text{ (обобщ. конст.)}$$

$$p_x = \text{const}$$

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial x} \right)^2 = 0$$

$$S = S(h, x, t)$$

$$S = S_0(h, t) + S_1(x, h)$$

$$\frac{\partial S_0}{\partial t} = -h \Rightarrow S_0 = -ht$$

$$\frac{\partial S_1}{\partial x} = p_x = \sqrt{2mh} \Rightarrow S_1 = \sqrt{2mh} x$$

$$S = -ht + \sqrt{2mh} x$$

$$\begin{cases} \frac{\partial S}{\partial x} = p_x \\ \frac{\partial S}{\partial h} = -\beta \end{cases}$$

$$p_x = \sqrt{2mh}$$

$$-t + \frac{x\sqrt{2m}}{\sqrt{h}} = -\beta$$

$$x = \frac{\sqrt{2h}}{\sqrt{m}} (t - \beta)$$

$$x(0) = x_0 \Rightarrow \beta \frac{\sqrt{2h}}{\sqrt{m}} = -x_0$$

$$x = \frac{\sqrt{2h}}{\sqrt{m}} t + x_0$$

$$p_x = m \dot{x}_0 = \sqrt{2mh} \Rightarrow \dot{x}_0 = \frac{\sqrt{2h}}{\sqrt{m}}$$

$$x = \dot{x}_0 t + x_0$$